Business Requirements Document (BRD)
Algorithmic Trading System

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Approval

Role	Name	Signature	Date
Product Owner	[To be assigned]		
Head of Trading	[To be assigned]		
Compliance Officer	[To be assigned]		

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1. Executive Summary

The Algorithmic Trading System (ATS) is a sophisticated, high-performance platform designed to execute paper and simulated trades across multiple asset classes, including stocks, indices, forex, and commodities, on major global exchanges such as NYSE, NASDAQ, LSE, and NSE. The system integrates with the Interactive Brokers (IBKR) Paper Trading API to facilitate simulated trading, leveraging advanced data processing libraries like Polars and DuckDB, and incorporates AI agents for enhanced decision-making. The ATS aims to provide traders with a robust, secure, and user-friendly environment to test and refine trading strategies, monitor market conditions, and analyze performance metrics in real time.

The primary objective of the ATS is to enable the hedge fund to develop, backtest, and deploy algorithmic trading strategies with precision and efficiency while adhering to strict risk management and regulatory compliance standards. The system will feature two core trading strategies: a Trend-Following Strategy and a Support and Resistance Strategy, both designed for long and short positions. A Market Watch Dashboard will serve as the primary user interface, providing real-time data visualization, technical analysis, portfolio monitoring, and simulated trading capabilities.

This Business Requirements Document (BRD) outlines the functional and non-functional requirements, integration needs, and implementation considerations for the ATS. It serves as a comprehensive guide for stakeholders, developers, and compliance teams to align on the system's objectives, scope, and deliverables. The development approach follows Agile Scrum with Test-Driven Development (TDD) and Continuous Integration/Continuous Deployment (CI/CD) practices to ensure rapid iteration, high quality, and adaptability to evolving market conditions.

The ATS is expected to deliver significant business value by enabling data-driven trading decisions, reducing operational risks, and providing a scalable platform for future enhancements, such as live trading and integration with alternative data sources. This document addresses all critical aspects of the system, including trading logic, user interface design, risk management protocols, and security measures, to ensure a robust and compliant solution.

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2. Introduction

2.1. Purpose

The purpose of this BRD is to define the business requirements for the development of the Algorithmic Trading System, a strategic initiative to enhance the hedge fund's trading capabilities through automation and advanced analytics. This document serves as a foundational reference for all stakeholders, including traders, developers, compliance officers, and management, to understand the system's objectives, scope, and requirements. It provides a detailed blueprint for the development team to design, implement, and test the system, ensuring alignment with business goals and regulatory standards.

2.2. Scope

The ATS will encompass the following key components:

- **Trading Strategies**: Implementation of two primary strategies—Trend-Following and Support and Resistance—supporting both long and short positions across stocks, indices, forex, and commodities.
- **Market Watch Dashboard**: A web-based, responsive user interface for real-time market monitoring, technical analysis, portfolio management, and simulated trading.

- **Data Management**: Efficient ingestion, processing, and storage of market data using Polars and DuckDB, with real-time and historical data support.
- **Order Execution**: Integration with the IBKR Paper Trading API for placing and managing simulated trades.
- **Risk Management**: Robust position sizing, stop-loss, take-profit, and portfolio-level risk controls to mitigate financial exposure.
- **AI Agents**: Modular AI components for market analysis, trade execution, and risk assessment, operating under human oversight.
- **Reporting and Analytics**: Comprehensive trade performance metrics, backtesting results, and portfolio analytics.
- **Security and Compliance**: Secure handling of sensitive data, encrypted communications, and adherence to financial regulations.

Out of Scope:

- Live trading functionality (to be considered in future phases).
- Integration with alternative brokers beyond IBKR.
- Support for cryptocurrencies or other non-traditional asset classes.
- Mobile app development (though the dashboard will be responsive for mobile devices).

2.3. Objectives

The ATS aims to achieve the following business objectives:

1. **Enhance Trading Efficiency**: Automate trading processes to reduce manual intervention, improve execution speed, and enable rapid response to market opportunities.
2. **Improve Decision-Making**: Provide traders with real-time data, technical indicators, and AI-driven insights to support informed trading decisions.
3. **Mitigate Risks**: Implement robust risk management protocols to limit financial exposure and ensure capital preservation.
4. **Ensure Compliance**: Adhere to regulatory requirements, including transaction reporting and data protection standards.
5. **Enable Strategy Testing**: Facilitate backtesting and paper trading to validate and optimize trading strategies without financial risk.
6. **Provide Scalability**: Build a modular and extensible platform to support future enhancements, such as additional strategies, asset classes, and live trading.
7. **Deliver User-Friendly Experience**: Create an intuitive and responsive Market Watch Dashboard to enhance trader productivity and engagement.

2.4. Assumptions and Constraints

Assumptions:

- The IBKR Paper Trading API will provide reliable and timely market data and order execution capabilities.
- The hedge fund has sufficient hardware resources (e.g., Lenovo Legion Pro 5 with NVIDIA RTX 3060 GPU) to support development and testing.
- Traders have basic familiarity with algorithmic trading concepts and technical indicators.
- Regulatory requirements for paper trading are minimal compared to live trading.
- The development team has expertise in Python, Polars, DuckDB, and web development frameworks like React and FastAPI.

Constraints:

- Development must be completed within a 3-6 month timeline to align with the hedge fund's strategic goals.
- The system must operate within IBKR API rate limits (e.g., 50 requests per second for market data).
- Budget constraints limit the use of premium third-party data sources or cloud infrastructure for initial deployment.
- The system must be deployed locally on the Lenovo Legion Pro 5, with no immediate cloud hosting.
- Compliance with financial regulations (e.g., SEC, GDPR) must be ensured, even for paper trading.

Assumptions Log (Appendix 11.1):

- Detailed assumptions and their impact on requirements are documented for stakeholder review.

2.5. Stakeholders

Stakeholder	Role	Responsibilities
Head of Trading	Oversees trading operations	Defines trading strategy requirements, approves deliverables
Traders	End-users of the ATS	Provide feedback on usability, test strategies
Product Owner	Represents business interests	Prioritizes features, manages backlog
Development Team	Builds and tests the system	Implements requirements, ensures quality
Compliance Officer	Ensures regulatory adherence	Reviews system for compliance, approves deployment
IT Administrator	Manages infrastructure	Configures hardware, ensures system uptime
Risk Manager	Oversees risk controls	Defines risk parameters, monitors performance

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3. Business Context

3.1. Business Need

The hedge fund seeks to modernize its trading operations by adopting algorithmic trading to enhance efficiency, reduce human error, and capitalize on market opportunities. The ATS addresses the following business needs:

- **Competitive Advantage**: Algorithmic trading enables faster and more precise trade execution compared to manual trading, positioning the hedge fund to outperform competitors.
- **Cost Reduction**: Automation reduces operational costs associated with manual trade monitoring and execution.
- **Risk Management**: Advanced risk controls protect capital and ensure sustainable returns, critical in volatile markets.
- **Strategy Validation**: Paper trading and backtesting capabilities allow the fund to test strategies without financial risk, improving confidence in deployment.
- **Data-Driven Insights**: Real-time market data and technical analysis empower traders to make informed decisions, enhancing portfolio performance.
- **Regulatory Compliance**: A compliant system ensures adherence to financial regulations, avoiding penalties and reputational risks.

3.2. Market Analysis

The global algorithmic trading market is projected to grow significantly, driven by advancements in technology, increasing market volatility, and the adoption of AI and machine learning in finance. Key trends include:

- **Rise of Retail and Institutional Algo Trading**: Both retail and institutional investors are adopting algorithmic trading for its efficiency and scalability.
- **Multi-Asset Trading**: Demand for platforms supporting diverse asset classes (stocks, forex, commodities) is increasing to diversify portfolios.
- **AI Integration**: AI-driven trading systems are gaining traction for their ability to analyze complex datasets and adapt to market changes.
- **Regulatory Scrutiny**: Regulators are imposing stricter requirements on algorithmic trading systems, necessitating robust compliance features.
- **Technology Adoption**: High-performance computing, cloud infrastructure, and advanced data analytics are becoming standard in trading platforms.

The ATS aligns with these trends by offering a multi-asset, AI-augmented platform with strong compliance and performance capabilities, positioning the hedge fund to capitalize on market opportunities.

3.3. Regulatory Considerations

The ATS must comply with relevant financial regulations, even in paper trading mode, to prepare for future live trading. Key regulations include:

- **SEC Regulations (U.S.)**: Transaction reporting, market abuse prevention, and data protection requirements under Regulation NMS and CAT.
- **GDPR (Europe)**: Protection of personal and financial data, with consent mechanisms for data processing.

- **MiFID II (Europe)**: Transparency in trade execution and reporting for EU markets.
- **SEBI (India)**: Compliance with algorithmic trading guidelines for NSE trades.
- **Data Security Standards**: Adherence to ISO 27001 and PCI DSS for secure data handling.

The system will implement audit trails, transaction logging, and compliance monitoring to meet these requirements. The Compliance Officer will review all features to ensure regulatory alignment.

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4. System Overview

4.1. System Description

The ATS is a comprehensive platform for algorithmic trading, designed to execute simulated trades using the IBKR Paper Trading API. It supports multiple asset classes (stocks, indices, forex, commodities) and integrates advanced data processing, AI agents, and a user-friendly Market Watch Dashboard. The system is built using a modular, event-driven architecture to ensure scalability, performance, and maintainability. Key components include:

- **Strategy Engine**: Executes Trend-Following and Support and Resistance strategies with configurable parameters.
- **Data Layer**: Processes and stores market data using Polars and DuckDB for high-performance analytics.
- **Order Execution Module**: Manages trade placement and monitoring via the IBKR API.
- **Risk Management Module**: Enforces position sizing, stop-loss, and portfolio-level risk controls.
- **AI Agents Framework**: Automates market analysis, trade execution, and risk assessment.
- **Market Watch Dashboard**: Provides real-time visualizations, technical analysis, and portfolio management.
- **API Layer**: Exposes system functionality via RESTful and WebSocket APIs for frontend and external integration.

The system is deployed locally on the Lenovo Legion Pro 5, leveraging its GPU and CPU for performance, and uses Docker for containerization to ensure consistency across environments.

4.2. Key Features

- **Multi-Asset Trading**: Supports stocks (NYSE, NASDAQ, LSE, NSE), indices (S&P 500, FTSE 100), forex (EUR/USD, USD/JPY), and commodities (gold, crude oil).
- **Real-Time Market Monitoring**: Displays live price updates, volume, and technical indicators via the Market Watch Dashboard.
- **Automated Trading Strategies**: Executes Trend-Following and Support and Resistance strategies with dynamic position sizing and risk management.
- **Interactive Dashboard**: Offers candlestick charts, technical indicator overlays, portfolio summaries, and simulated trading controls.
- **AI-Driven Insights**: Provides market analysis, risk assessments, and trade execution recommendations through modular AI agents.
- **Backtesting and Simulation**: Validates strategies against historical data and simulates trades in the IBKR paper trading environment.
- **Robust Security**: Implements encrypted communications, secure credential management, and role-based access control.
- **Comprehensive Reporting**: Generates trade performance metrics, portfolio analytics, and compliance reports.

4.3. System Architecture Principles

The ATS adheres to the following architectural principles:

- **Modularity**: Components (e.g., strategy engine, data layer) are loosely coupled, enabling independent development and scalability.
- **Event-Driven Design**: Market events trigger strategy calculations, risk assessments, and order executions for low-latency responses.
- **Separation of Concerns**: Trading logic, risk management, and order execution are isolated to enhance maintainability.
- **Data-Centric Approach**: Efficient data pipelines ensure high-quality market data for trading decisions.
- **Security by Design**: Encryption, authentication, and vulnerability protections are embedded at every layer.

- **Fault Tolerance**: Automatic reconnections, error handling, and circuit breakers ensure operational continuity.
- **Observability**: Structured logging, performance metrics, and health monitoring provide real-time system insights.

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5. Functional Requirements

5.1. Trading Strategies

The ATS will implement two core trading strategies: Trend-Following and Support and Resistance, both supporting long and short positions. These strategies are designed for positional trading and will be executed in the IBKR paper trading environment.

5.1.1. Trend-Following Strategy

Description: A positional strategy that captures long-term price movements using volume-weighted technical indicators to identify trends, momentum, and volatility.
Assets: Stocks, indices, forex, commodities.
Technical Indicators:

- 55-day Volume-Weighted Simple Moving Average (VW SMA) of Open: Identifies long-term trend direction.
- 13-day VW SMA of Open: Tracks medium-term price movements (optional).
- 14-day Volume-Weighted Money Flow Index (MFI) of HLC Average: Measures momentum.
- 34-day VW SMA of MFI(14): Smooths MFI for trend confirmation.
- 21-day VW SMA of MFI(14): Triggers exits.
- 5-day VW SMA of High: Defines short-term resistance levels.
- 5-day VW SMA of Low: Defines short-term support levels.
- 21-day Volume-Weighted Average True Range (ATR): Measures volatility for stop-loss and take-profit.

Entry Rules:

- **Long Position**:
  - 14-day VW MFI > 34-day VW SMA of MFI(14).
  - Current price > 55-day VW SMA of Open.
  - Action: Place buy order via IBKR API.
- **Short Position**:
  - 14-day VW MFI < 34-day VW SMA of MFI(14).
  - Current price < 55-day VW SMA of Open.
  - Action: Place sell order via IBKR API.

Exit Rules:

- **Long Position**:
  - Stop-Loss: 2x 21-day VW ATR below entry price.
  - Take-Profit: 2x 21-day VW ATR above entry price.
  - Additional Exits:
    - 14-day VW MFI < 21-day VW SMA of MFI(14).
    - Current price < 5-day VW SMA of Low.
- **Short Position**:
  - Stop-Loss: 2x 21-day VW ATR above entry price.
  - Take-Profit: 2x 21-day VW ATR below entry price.
  - Additional Exits:
    - 14-day VW MFI > 21-day VW SMA of MFI(14).
    - Current price > 5-day VW SMA of High.

Position Sizing:

- Risk per trade: 1% of account balance (£10,000 initial balance).
- Formula:

$$\text{Position Size} = \frac{\text{Account Balance} \times 0.01}{2 \times \text{21-day VW ATR} \times \text{Lot Size}}$$

- Lot Size:

    - Stocks: 1 share.
    - Forex: 100,000 units.
    - Commodities: Per contract (e.g., 100 oz for gold).
    - Indices: Per contract multiplier (e.g., 10 for S&P 500).

Money Management:

- Maximum Drawdown: If drawdown > 20%, reduce risk to 0.5% per trade.
- Trailing Stop: 1.5x 21-day VW ATR below highest price (long) or above lowest price (short).
- Currency Adjustments: Convert risk calculations to account base currency (GBP).

5.1.2. Support and Resistance Strategy

Description: A positional strategy that trades breakouts and reversals at key support and resistance levels using Gann Fan, Fibonacci retracement, and Pivot Points.

Assets: Stocks, indices, forex, commodities.

Technical Tools:

- **Pivot Points**:
    - Formulas:

$$\text{PP} = \frac{\text{High} + \text{Low} + \text{Close}}{3}$$

$$\text{R1} = 2 \times \text{PP} - \text{Low}$$

$$\text{S1} = 2 \times \text{PP} - \text{High}$$

$$\text{R2} = \text{PP} + (\text{High} - \text{Low})$$

$$\text{S2} = \text{PP} - (\text{High} - \text{Low})$$

- **Fibonacci Retracement**: Levels at 23.6%, 38.2%, 50%, 61.8%, 100% based on recent price swings.
- **Gann Fan**: Angles (1x1, 1x2, 2x1) from significant highs/lows for dynamic support/resistance.

Entry Rules:

- **Long Position**:
    - Price breaks above R1 or a Fibonacci level (e.g., 61.8%).
    - Price bounces off a Gann Fan angle (e.g., 1x1) in an uptrend.
    - Optional: Bullish candlestick pattern or rising volume.
    - Action: Place buy order via IBKR API.
- **Short Position**:
    - Price breaks below S1 or a Fibonacci level (e.g., 38.2%).
    - Price rejected at a Gann Fan angle (e.g., 1x1) in a downtrend.
    - Optional: Bearish candlestick pattern or rising volume.
    - Action: Place sell order via IBKR API.

Exit Rules:

- **Long Position**:
    - Take-Profit: At R2, next Fibonacci level, or next Gann Fan angle.
    - Stop-Loss: Below S1 or recent swing low (or 1-2% below entry).
- **Short Position**:
    - Take-Profit: At S2, next Fibonacci level, or next Gann Fan angle.
    - Stop-Loss: Above R1 or recent swing high (or 1-2% above entry).
- **Trailing Stop**:
    - Percentage-Based: 2-3% below highest price (long) or above lowest price (short).
    - Volatility-Based: 2x 14-day ATR.

Position Sizing:

- Risk per trade: 1-2% of account balance.

- Formula:
  [
  \text{Position Size} = \frac{\text{Account Balance} \times \text{Risk %}}{\text{|Entry Price - Stop Loss Price|}}
  ]

Money Management:

- Total Portfolio Risk: <5-10% of account balance across all trades.
- Diversification: Avoid concentration in a single asset or sector.
- Volatility Adjustment: Use ATR to scale position sizes.

5.2. Market Watch Dashboard

Description: A web-based, responsive dashboard for real-time market monitoring, technical analysis, portfolio management, and simulated trading.

Features:

- **Scrip Selection**: Dropdown or input field to select tickers (e.g., AAPL, EUR/USD).
- **Real-Time Data Display**:
  - Bid, ask, last price, volume, and OHLCV data in a table or ticker format.
  - Updated every 5-10 seconds via IBKR API.
- **Interactive Charts**:
  - Candlestick or line charts with zoom/pan capabilities.
  - Overlay technical indicators (VW SMA, MFI, ATR, RSI, MACD, Gann Fan, Pivot Points, Fibonacci).
  - Adjustable timeframes (1m, 5m, 1h, 1d, 1w).
- **Portfolio Summary**:
  - Positions, account value, unrealized P&L for paper trading account.
  - Filterable and sortable tables.
- **Simulated Trading Panel**:
  - Place buy/sell orders (market, limit, stop) via IBKR API.
  - Order entry form with validation (ticker, quantity, price).
  - Real-time order status updates.
- **Watchlist**:
  - Save and monitor favorite scrips.
  - Highlight high-volatility and high-volume scrips.
- **Technical Analysis**:
  - Daily and weekly outlook (Strong Uptrend, Uptrend, Sideways, Downtrend, Strong Downtrend).
  - Indicator status (RSI(14), MFI(14), MACD, SMAs, EMAs).
  - Support and resistance levels with trading recommendations.
- **News and Events**:
  - Latest news articles for selected scrips.
  - Upcoming events (AGM, dividends) within 10 days.
- **Responsiveness**: Optimized for desktop, tablet, and mobile devices (iPhone, Android).

Implementation:

- Frontend: React.js with Tailwind CSS and Plotly.js.
- Backend: FastAPI for API endpoints, ib_insync for IBKR integration.
- Data Processing: Polars for indicator calculations, DuckDB for storage.

5.3. Data Management

Description: Efficient ingestion, processing, storage, and retrieval of market data to support trading and analysis.

Requirements:

- **Data Sources**:
  - Real-time and historical OHLCV data from IBKR Paper Trading API.
  - Future integration with news feeds and sentiment data (out of scope for initial phase).
- **Data Ingestion**:
  - Fetch data via ib_insync with WebSocket for real-time updates.
  - Poll IBKR every 5-10 seconds, respecting API rate limits.

    - Cache data in Redis for low-latency access.
- **Data Processing**:
    - Normalize data formats (timestamps, asset identifiers).
    - Validate and clean data to remove outliers and missing values.
    - Compute technical indicators using Polars for high performance.
- **Data Storage**:
    - DuckDB with Parquet for time-series data (market data, trades, positions).
    - Partition by date and asset for efficient queries.
    - Daily backups to external storage.
- **Data Visualization**:
    - Render charts with Plotly.js for price, indicators, and performance metrics.
    - Support multi-timeframe analysis and export to PNG/CSV.

5.4. Order Execution

Description: Manages the trade lifecycle, from order creation to execution and monitoring, in the IBKR paper trading environment.

Requirements:

- **Order Types**: Market, limit, stop, and bracket orders (entry, take-profit, stop-loss).
- **Order Placement**:
    - Submit orders via ib_insync with validation (price, quantity, asset).
    - Support partial fills and cancellations.
- **Order Monitoring**:
    - Track order status (pending, filled, canceled) in real time.
    - Update portfolio and P&L upon execution.
- **Error Handling**:
    - Handle API connectivity issues with automatic reconnections.
    - Log failed orders for analysis.
- **Simulation Mode**: Ensure all trades are executed in paper trading mode to prevent live trades.

5.5. Risk Management

Description: Enforces risk controls to protect capital and ensure sustainable trading.

Requirements:

- **Position Sizing**:
    - Limit risk to 1-2% per trade based on account balance and volatility (ATR).
    - Adjust for lot size and asset type.
- **Stop-Loss and Take-Profit**:
    - Set stop-loss and take-profit levels based on ATR or support/resistance levels.
    - Place orders with IBKR API upon entry.
- **Trailing Stop**:
    - Implement trailing stops (1.5x ATR for Trend-Following, 2-3% for Support and Resistance).
    - Update via IBKR API as price moves.
- **Portfolio Risk**:
    - Limit total open risk to 5-10% of account balance.
    - Diversify across assets and sectors.
    - Monitor correlations to avoid concentrated exposure.
- **Drawdown Limit**:
    - Reduce risk to 0.5% per trade if drawdown > 20%.
- **Circuit Breakers**:
    - Halt trading during extreme volatility or system anomalies.
    - Trigger alerts for manual review.
- **Currency Handling**:
    - Convert risk calculations to GBP for consistency.

5.6. AI Agents Integration

Description: Modular AI agents automate trading tasks, including market analysis, trade execution, and risk assessment.

Requirements:

- **Core Agents**:
  - **Trade Execution Agent**: Places orders based on strategy signals, optimizing for price and timing.
  - **Risk Management Agent**: Enforces position sizing, exposure limits, and circuit breakers.
  - **Market Analysis Agent**: Generates insights on trends, volatility, and momentum.
  - **IBKR API Agent**: Manages API interactions for data and orders.
- **Implementation**:
  - Use Python with scikit-learn and TensorFlow for agent logic.
  - RabbitMQ for event-driven communication.
  - FastAPI for agent APIs.
- **Transparency**:
  - Log agent decisions for auditability.
  - Provide explainability using SHAP or LIME.
- **Human Oversight**:
  - Require human approval for critical decisions.
  - Allow pausing or overriding agents via the dashboard.
- **Testing**:
  - Validate agents through backtesting and paper trading.
  - Ensure adherence to risk parameters.

5.7. User Interface

Description: A responsive, intuitive web interface for traders to monitor markets, analyze data, and manage trades.

Requirements:

- **Frontend Framework**: React.js with Tailwind CSS for styling.
- **Components**:
  - Dropdown/input for scrip selection.
  - Real-time price and volume tables.
  - Interactive candlestick/line charts with Plotly.js.
  - Portfolio and order status tables.
  - Simulated trading form with validation.
  - Watchlist with volatility and volume highlights.
  - Technical analysis panel with outlook and indicator status.
  - News and events section.
- **Interactivity**:
  - Dynamic updates via WebSocket for real-time data.
  - Callbacks for user inputs (e.g., changing scrips, submitting orders).
- **Responsiveness**:
  - Optimized for desktop, tablet, and mobile devices.
  - Support iPhone and Android browsers.
- **Accessibility**:
  - Adhere to WCAG 2.1 guidelines for color contrast and navigation.

5.8. Reporting and Analytics

Description: Generates performance metrics, trade logs, and compliance reports for analysis and auditing.

Requirements:

- **Trade Performance**:
  - Metrics: P&L, win rate, Sharpe ratio, max drawdown, annualized return.
  - Visualize equity curves and drawdowns with Plotly.js.
- **Portfolio Analytics**:
  - Exposure, correlations, and risk metrics (VaR, expected shortfall).
  - Diversification analysis across assets and sectors.
- **Backtesting Reports**:
  - Compare strategy performance against benchmarks.
  - Include sensitivity analysis for parameters.
- **Compliance Reports**:
  - Transaction logs with timestamps and user IDs.

- Audit trails for regulatory reporting.
- **Export Capabilities**:
  - Export reports as CSV, PDF, or PNG.
  - Schedule automated report generation.

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6. Non-Functional Requirements

6.1. Performance

- **Latency**: End-to-end latency from data ingestion to order submission <100ms under normal conditions.
- **Throughput**: Handle 10,000 market data updates per second and 1,000 trades per minute.
- **Indicator Calculation**: Compute technical indicators for 100 scrips in <1 second using Polars.
- **Dashboard Updates**: Refresh real-time data every 5-10 seconds without UI lag.
- **Backtesting**: Process 5 years of daily data for 10 scrips in <10 minutes.

6.2. Scalability

- **Vertical Scaling**: Leverage Lenovo Legion Pro 5's GPU and CPU for initial scaling.
- **Horizontal Scaling**: Support microservices architecture for future cloud deployment.
- **Data Scalability**: Handle 1TB of historical data with DuckDB and Parquet.
- **User Scalability**: Support up to 10 concurrent users on the dashboard.

6.3. Security

- **Authentication**: OAuth2 with JWT and MFA for administrative access.
- **Data Protection**: TLS 1.3 for all communications, AES-256 encryption for data at rest.
- **Credential Management**: Store IBKR API keys in an encrypted vault (e.g., HashiCorp Vault).
- **API Security**: Rate limiting, CSRF tokens, and input validation to prevent attacks.
- **Agent Security**: Sandbox AI agents in isolated containers with restricted permissions.

6.4. Reliability and Fault Tolerance

- **Uptime**: Achieve 99.9% uptime, excluding scheduled maintenance.
- **Error Handling**: Automatically reconnect to IBKR API after connectivity issues.
- **Data Integrity**: Validate and clean market data to ensure accuracy.
- **Circuit Breakers**: Halt trading during anomalies, with alerts to administrators.
- **Backup and Recovery**: Daily backups with recovery time objective (RTO) <1 hour.

6.5. Usability

- **Intuitive Design**: Dashboard navigation requires <5 minutes of training.
- **Response Time**: UI actions (e.g., chart zooming, order submission) complete in <1 second.
- **Accessibility**: Comply with WCAG 2.1 for color contrast and keyboard navigation.
- **Documentation**: Provide user guides and developer documentation in Confluence.

6.6. Compliance

- **Regulatory Adherence**: Comply with SEC, GDPR, MiFID II, and SEBI requirements.
- **Audit Trails**: Log all trades, orders, and system actions with immutable records.
- **Data Retention**: Store transaction logs for 7 years per regulatory guidelines.
- **Compliance Monitoring**: Real-time checks for market abuse (e.g., spoofing, wash trades).

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7. Integration Requirements

7.1. Interactive Brokers Paper Trading API

Description: The primary data source and execution platform for the ATS.

Requirements:

- **Connection Details**:
  - Host: 127.0.0.1
  - Port: 7497
  - Client ID: 0
  - Account No.: DUK221396
- **Data Fetching**:
  - Real-time OHLCV data via WebSocket.
  - Historical data for backtesting (up to 5 years).
- **Order Execution**:
  - Support market, limit, stop, and bracket orders.

- Monitor order status and handle partial fills.
- **Rate Limits**:
  - Respect IBKR limits (50 requests/second for market data, 100 messages/second for orders).
  - Implement caching to reduce API calls.
- **Error Handling**:
  - Automatic reconnection after timeouts or disconnects.
  - Log API errors for debugging.
- **Security**:
  - Securely store API credentials using environment variables or encrypted vaults.

7.2. External Data Sources

Description: Future integration with alternative data sources to enhance analysis (out of scope for initial phase).

Requirements:

- **News Feeds**: Integrate Reuters or Bloomberg for real-time news and events.
- **Sentiment Data**: Process social media or news sentiment for market insights.
- **Economic Indicators**: Incorporate GDP, inflation, or employment data for macro analysis.
- **Integration Patterns**:
  - RESTful APIs or WebSocket for real-time data.
  - Batch processing for historical data.
  - Normalize data formats for consistency.

7.3. Internal Systems

Description: Integration with internal systems for reporting and compliance.

Requirements:

- **Portfolio Management**: Sync with internal portfolio tools for benchmarking.
- **Compliance Systems**: Export transaction logs to compliance platforms.
- **Data Warehouse**: Store historical data in a centralized repository for analytics.
- **APIs**: Use RESTful APIs for data exchange, with OAuth2 authentication.

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8. Implementation Considerations

8.1. Development Methodology

- **Agile Scrum**: Two-week sprints with planning, daily stand-ups, reviews, and retrospectives.
- **Test-Driven Development (TDD)**: Write tests before code for strategies, risk management, and order execution.
- **CI/CD**: Use GitHub Actions for automated builds, tests, and deployments.
- **Roles**:
  - Product Owner: Prioritizes features and validates deliverables.
  - Scrum Master: Facilitates Agile processes and removes impediments.
  - Development Team: Implements and tests the system.

8.2. Technology Stack

- **Backend**: Python 3.10, FastAPI, ib_insync.
- **Data Processing**: Polars for analytics, DuckDB with Parquet for storage.
- **Frontend**: React 18, Tailwind CSS, Plotly.js.
- **Infrastructure**: Docker, Docker Compose, GitHub Actions.
- **Monitoring**: Prometheus, Grafana, Python logging.
- **AI Agents**: scikit-learn, TensorFlow, RabbitMQ.

8.3. Testing Strategy

- **Unit Testing**: Validate individual components (indicators, strategies, agents) with pytest.
- **Integration Testing**: Verify module interactions (data to strategy, strategy to execution).
- **System Testing**: Simulate end-to-end trading workflows in staging.
- **Performance Testing**: Measure latency and throughput with Locust.
- **Security Testing**: Scan for vulnerabilities with OWASP ZAP.
- **Backtesting**: Test strategies against historical data.
- **Paper Trading**: Validate functionality in IBKR paper trading environment.
- **Coverage**: Achieve >80% code coverage.

8.4. Deployment Strategy

- **Local Deployment**: Deploy on Lenovo Legion Pro 5 using Docker Compose.
- **Containerization**: Dockerfiles for backend, frontend, database, and agents.
- **CI/CD Pipeline**:
  - Build and test on every commit.
  - Deploy to test environment automatically, production with manual approval.
- **Monitoring**: Prometheus and Grafana for real-time metrics.
- **Backup**: Daily backups of DuckDB and Parquet data.
- **Rollback**: Automated rollback to previous version if deployment fails.

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9. Risks and Mitigation

Risk Category	Description	Likelihood	Impact	Mitigation Strategy
Market Risk	Volatility affects strategy performance	High	High	Implement circuit breakers, diversify assets
Operational Risk	System failures disrupt trading	Medium	High	Automatic reconnections, comprehensive error handling
Model Risk	Strategy or AI errors lead to losses	Medium	High	Backtesting, paper trading validation, human oversight
Regulatory Risk	Non-compliance with SEC, GDPR, etc.	Low	High	Compliance monitoring, audit trails, external audits
Cybersecurity Risk	Data breaches or unauthorized access	Low	High	Encryption, authentication, regular security testing
API Rate Limits	IBKR API restrictions limit data access	Medium	Medium	Caching, optimized API calls, rate limit monitoring
Resource Constraints	Hardware or budget limitations	Low	Medium	Optimize resource usage, prioritize critical features

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10. Glossary

- **ATS**: Algorithmic Trading System.
- **IBKR**: Interactive Brokers.
- **VW SMA**: Volume-Weighted Simple Moving Average.
- **MFI**: Money Flow Index.
- **ATR**: Average True Range.
- **Pivot Points**: Calculated support and resistance levels.
- **Fibonacci Retracement**: Percentage-based retracement levels.
- **Gann Fan**: Dynamic support/resistance angles.
- **Polars**: High-performance data processing library.
- **DuckDB**: In-memory analytical database.
- **FastAPI**: Python framework for building APIs.
- **ib_insync**: Python library for IBKR API integration.

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11. Appendices

11.1. Assumptions Log

Assumption ID	Description	Impact if Invalid	Validation Plan
A001	IBKR API provides reliable data	Delayed or inaccurate trades	Test API connectivity and data quality
A002	Traders have basic algo trading knowledge	Increased training needs	Conduct user training sessions
A003	Hardware supports system requirements	Performance bottlenecks	Benchmark hardware performance

Assumption ID	Description	Impact if Invalid	Validation Plan
A004	Minimal regulatory requirements for paper trading	Compliance delays	Consult Compliance Officer

11.2. Missing Information and Best Practices

Missing Information:

- Specific formulas for volume-weighted indicators (VW SMA, VW MFI, VW ATR).
- Detailed user personas for traders (e.g., experience level, preferences).
- Exact regulatory requirements for paper trading across jurisdictions.
- Performance benchmarks for similar trading systems.
- Preferred news feed providers for scrip-specific news.

Best Practices Applied:

- **Indicator Formulas**: Assumed standard volume-weighted calculations (e.g., VW SMA = Σ(Price $\times$ Volume) / ΣVolume). Reason: Industry-standard approach ensures compatibility and accuracy.
- **User Personas**: Assumed traders with intermediate knowledge of technical analysis. Reason: Aligns with typical hedge fund trader profiles, allowing focus on usability.
- **Regulatory Compliance**: Implemented comprehensive audit trails and transaction logging. Reason: Prepares system for live trading and covers common regulatory needs.
- **Performance Benchmarks**: Targeted <100ms latency and 10,000 updates/second throughput. Reason: Based on high-frequency trading standards, ensuring competitiveness.
- **News Feeds**: Planned for future integration with Reuters or Bloomberg. Reason: Widely used in financial industry for reliable, timely data.

11.3. References

- Interactive Brokers API Documentation: https://interactivebrokers.github.io/tws-api/
- Polars Documentation: https://pola.rs/
- DuckDB Documentation: https://duckdb.org/
- FastAPI Documentation: https://fastapi.tiangolo.com/
- React Documentation: https://reactjs.org/
- Plotly.js Documentation: https://plotly.com/javascript/
- SEC Regulations: https://www.sec.gov/
- GDPR Guidelines: https://gdpr.eu/